

HOMEWORK 2, WEEK 3

$$T_{\alpha_1, \alpha_2}(v_1, v_2) = \det \begin{bmatrix} \alpha_1(v_1) & \alpha_1(v_2) \\ \alpha_2(v_1) & \alpha_2(v_2) \end{bmatrix} : V \times V \rightarrow \mathbb{R}$$
$$= \alpha_1(v_1)\alpha_2(v_2) - \alpha_1(v_2)\alpha_2(v_1) \quad *$$

① $(0,2)$ -tensor = bilinear map

· linearity in first slot $T(av_1 + bv_2, u) = \begin{vmatrix} \alpha_1(av_1 + bv_2) & \alpha_1(u) \\ \alpha_2(av_1 + bv_2) & \alpha_2(u) \end{vmatrix} =$

$$= \begin{vmatrix} a\alpha_1(v_1) + b\alpha_1(v_2) & \alpha_1(u) \\ a\alpha_2(v_1) + b\alpha_2(v_2) & \alpha_2(u) \end{vmatrix} \quad \text{since } \alpha \text{ linear forms} =$$

$$= a \begin{vmatrix} \alpha_1(v_1) & \alpha_1(u) \\ \alpha_2(v_1) & \alpha_2(u) \end{vmatrix} + b \begin{vmatrix} \alpha_1(v_2) & \alpha_1(u) \\ \alpha_2(v_2) & \alpha_2(u) \end{vmatrix} \quad \text{column linearity} =$$

$$= aT(v_1, u) + bT(v_2, u)$$

· linearity in second slot follows from the first slot and antisymmetric property of determinant

② Let $\beta = \{b_1, \dots, b_n\}$ basis of V and $\beta^* = \{\beta_1, \dots, \beta_n\}$ basis of V^* .

Then $\beta_k(b_i) = \delta_{ki}$ ↘

$$\text{and } T_{\beta_k \beta_l}(b_i, b_j) = \delta_{ki}\delta_{lj} - \delta_{kj}\delta_{li} \in \{-1, 0, 1\}$$
$$= \begin{cases} 1, & \text{if } (i, j) = (k, l) \wedge i \neq j \\ -1, & \text{if } (i, j) = (l, k) \wedge i \neq j \\ 0, & \text{otherwise} \end{cases}$$

$$\textcircled{3} T_{\beta_k \beta_1}(v_1, v_2) = \beta_k(v_1) \beta_1(v_2) - \beta_k(v_2) \beta_1(v_1) \quad *$$

$$\begin{aligned} (\beta_k \otimes \beta_1 - \beta_1 \otimes \beta_k)(v_1, v_2) &= (\beta_k \otimes \beta_1)(v_1, v_2) - (\beta_1 \otimes \beta_k)(v_1, v_2) = \\ &= \beta_k(v_1) \beta_1(v_2) - \beta_k(v_2) \beta_1(v_1) \text{ by definition} \end{aligned}$$

therefore identical

STATEMENT: This is my work. For this part I did not use Gemini.